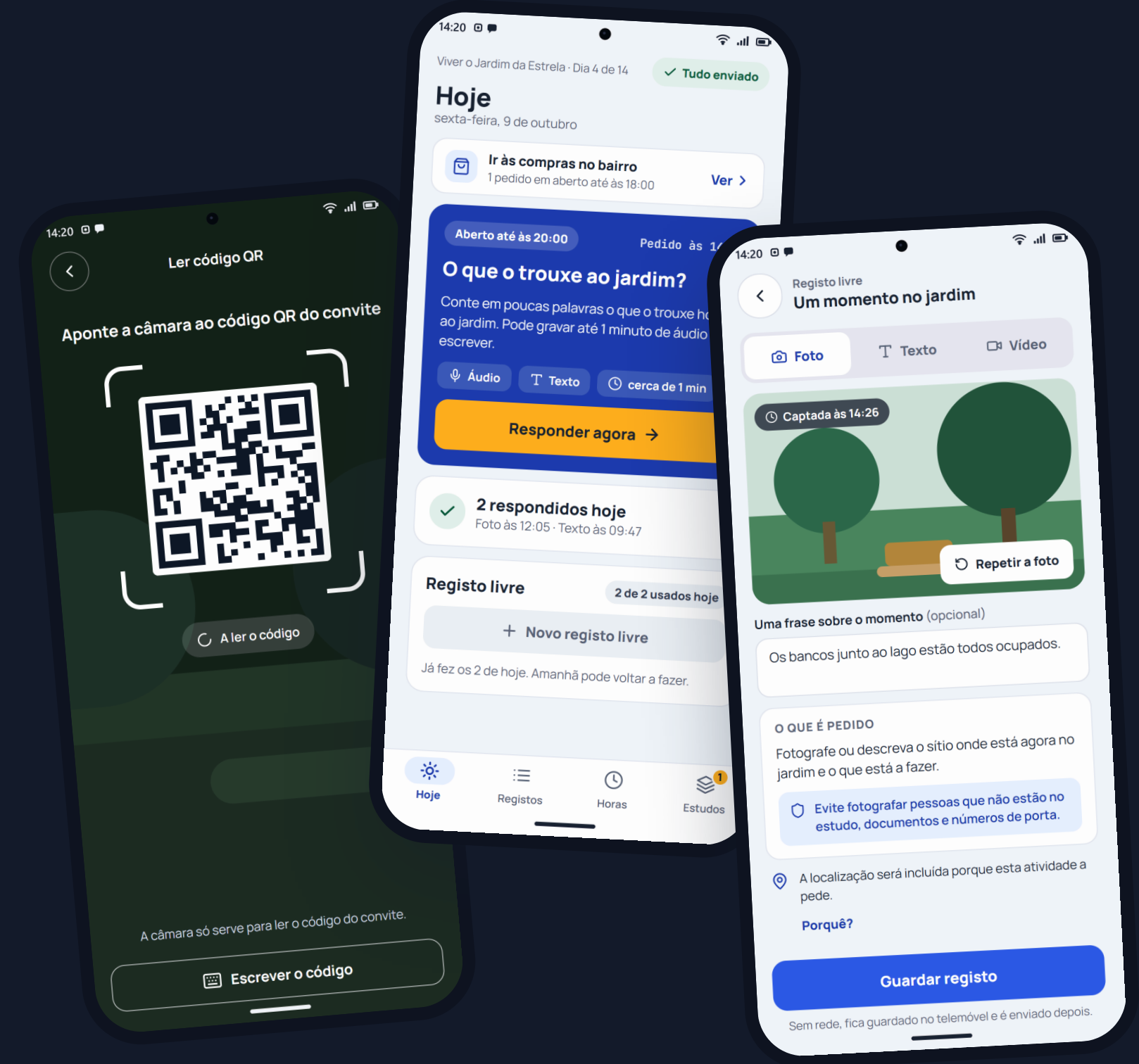




Record life as it happens

A distributed, fault-tolerant platform for mobile ethnography and diary studies

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BSc in Computer Engineering · IADE, Universidade Europeia



THE PROBLEM

An entry made later is no longer the same

Out in the street, offline

The entry has to stay saved on the phone.

Everyone at once

A prompt sent to everyone brings a burst of photos and videos.

PAPER DIARY

Participants said

90%

Actual adherence

11%

Stone et al., BMJ, 2002

WHO IT IS FOR

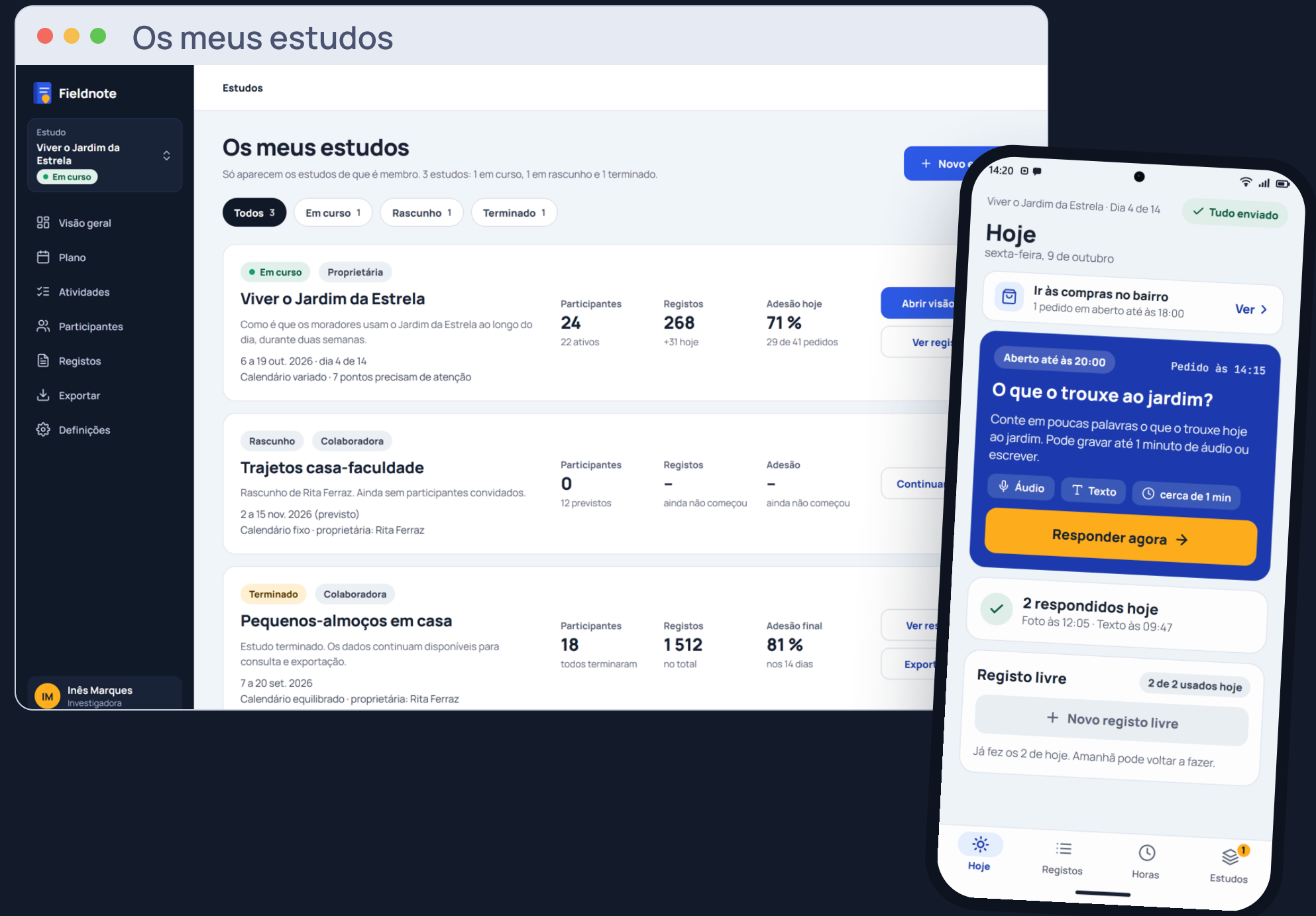
Research in design and communication

The framing is UNIDCOM/IADE, the IADE research unit.

Data under the institution's control

Photo, video, audio, text and scale

App and consent in Portuguese



WHAT ALREADY EXISTS

Each feature exists elsewhere

PLATFORM	TYPE	HOSTED BY	FOCUS
Indeemo, dscout, EthOS	Commercial	The vendor	Rich media, recruitment
m-Path	Academic, closed	The vendor (Azure, Frankfurt)	EMA questionnaires
Beiwe, AWARE	Open source	Whoever installs it (Beiwe only on AWS)	Phone sensors
Fieldnote	Academic project	The institution, on its own server	Rich media, fault tolerance, CSP scheduling

What sets Fieldnote apart: the institution hosts the data.

THE SOLUTION

One system, three roles

ANDROID APP

Participant

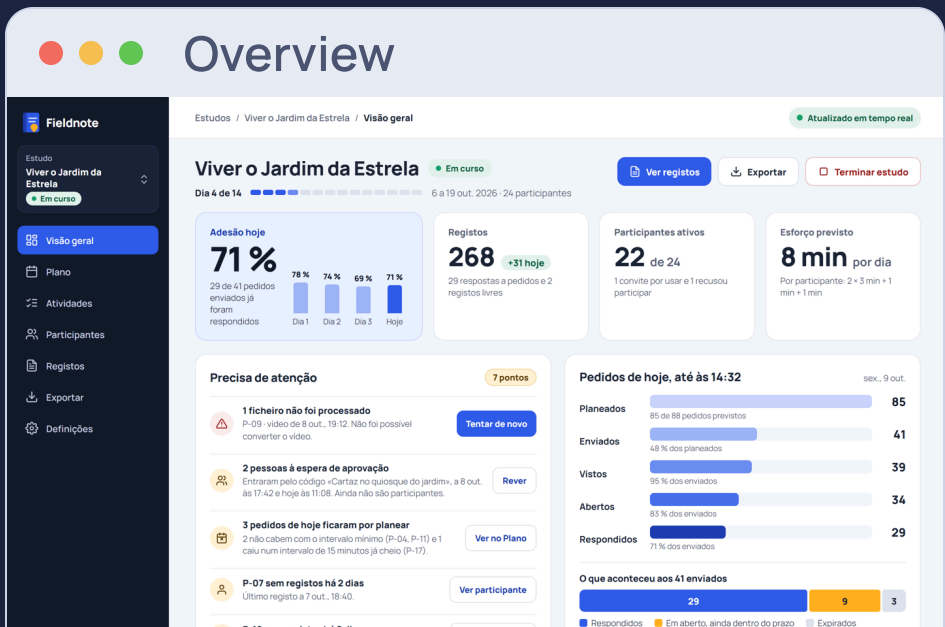
Joins a study, receives prompts and records entries.



WEB PORTAL

Researcher

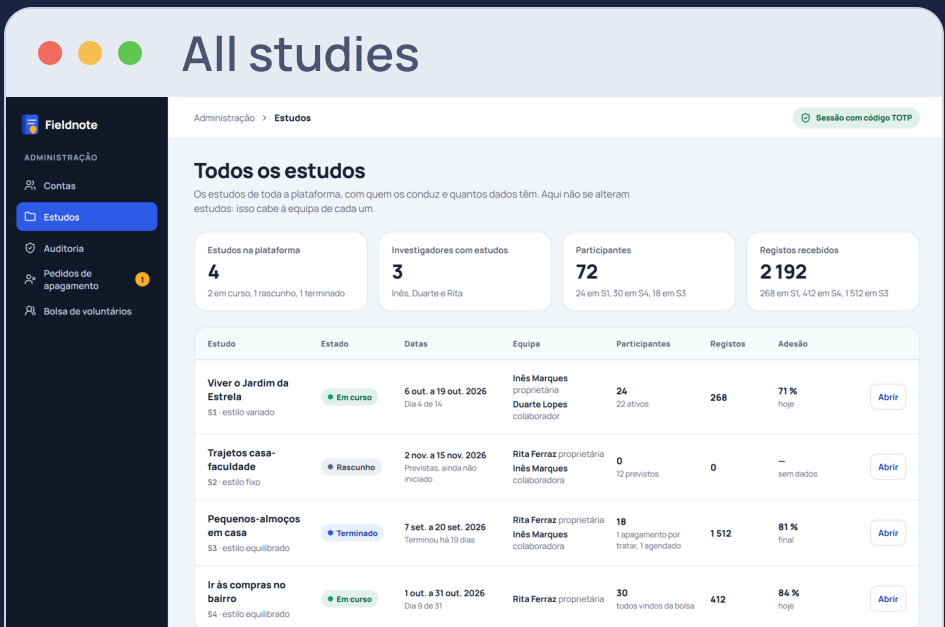
Creates the study and follows it live.



WEB PORTAL

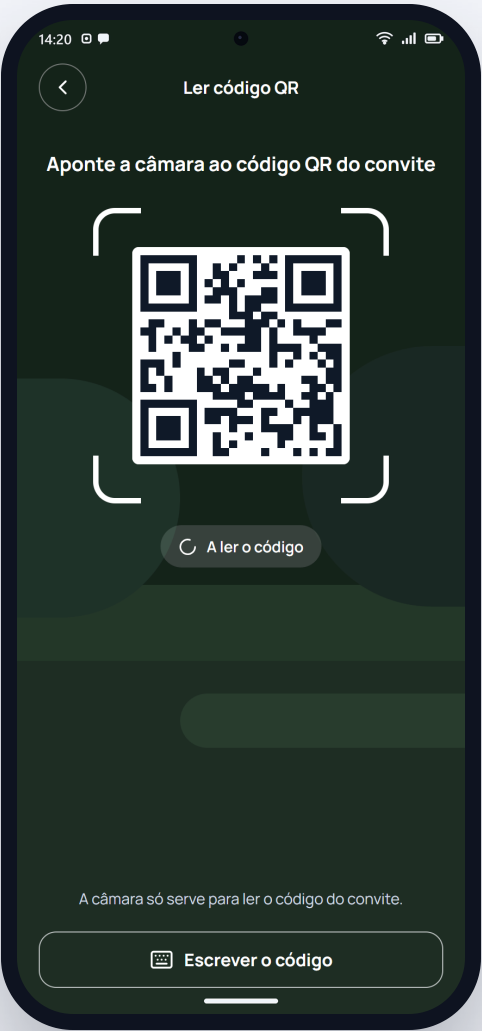
Administrator

Manages accounts, studies and deletions.



PARTICIPANT

Three ways to join a study



Personal invite code
Scanned as a QR



Or typed
12 characters, single use



Volunteer pool
Invitation in the app



Poster
8 characters, reusable

PARTICIPANT

One prompt, one tap, one entry

The notification arrives, the app opens on Today and the participant chooses how to respond.

Text

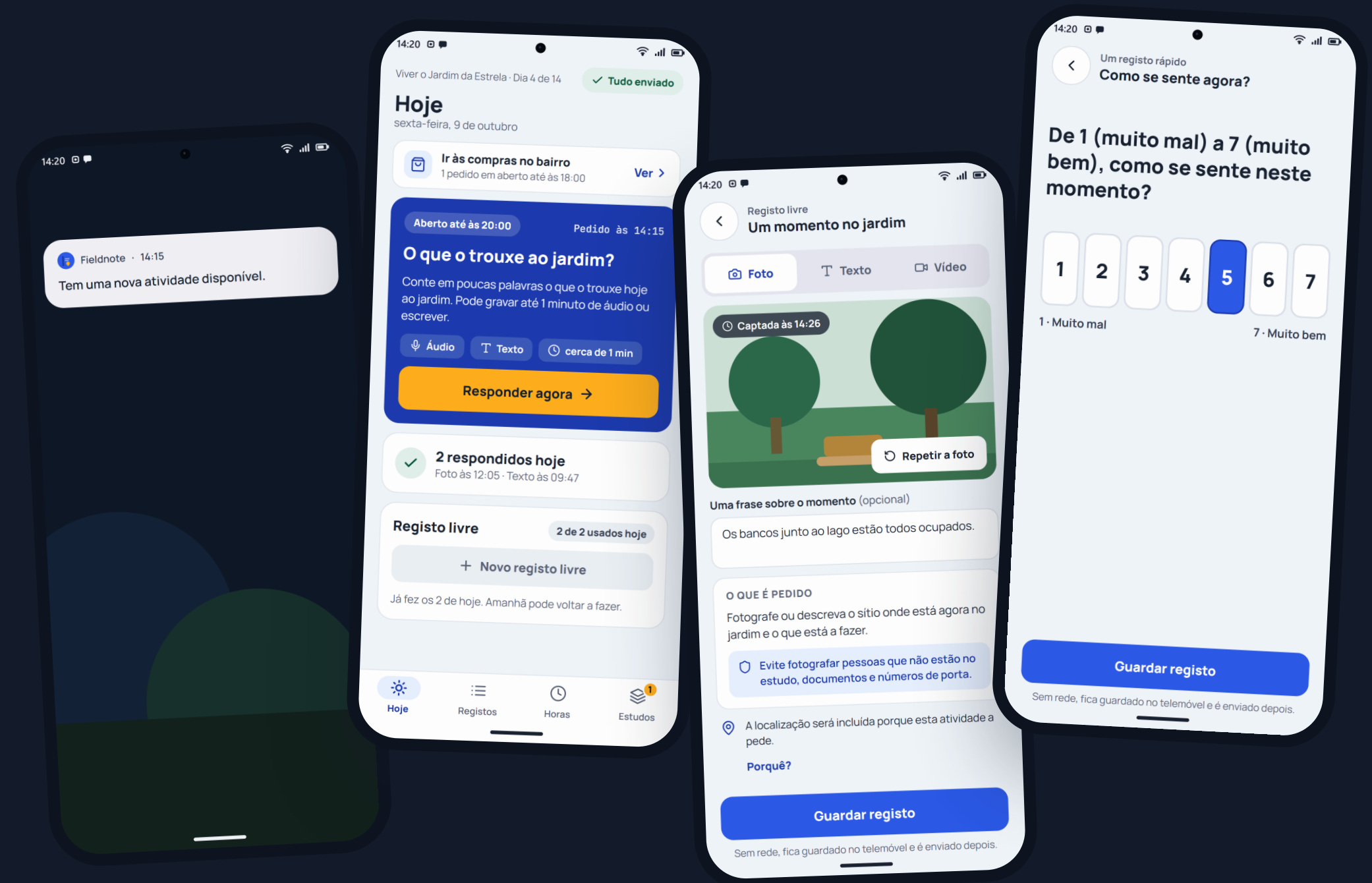
Photo

Video

Audio

Scale from 1 to 7

Multiple choice



PARTICIPANT

Without a network, the entry is kept

The phone keeps it in a local queue and resends when there is a connection. Resending does not duplicate.



PARTICIPANT

The participant is in control

- 1

Consent before starting

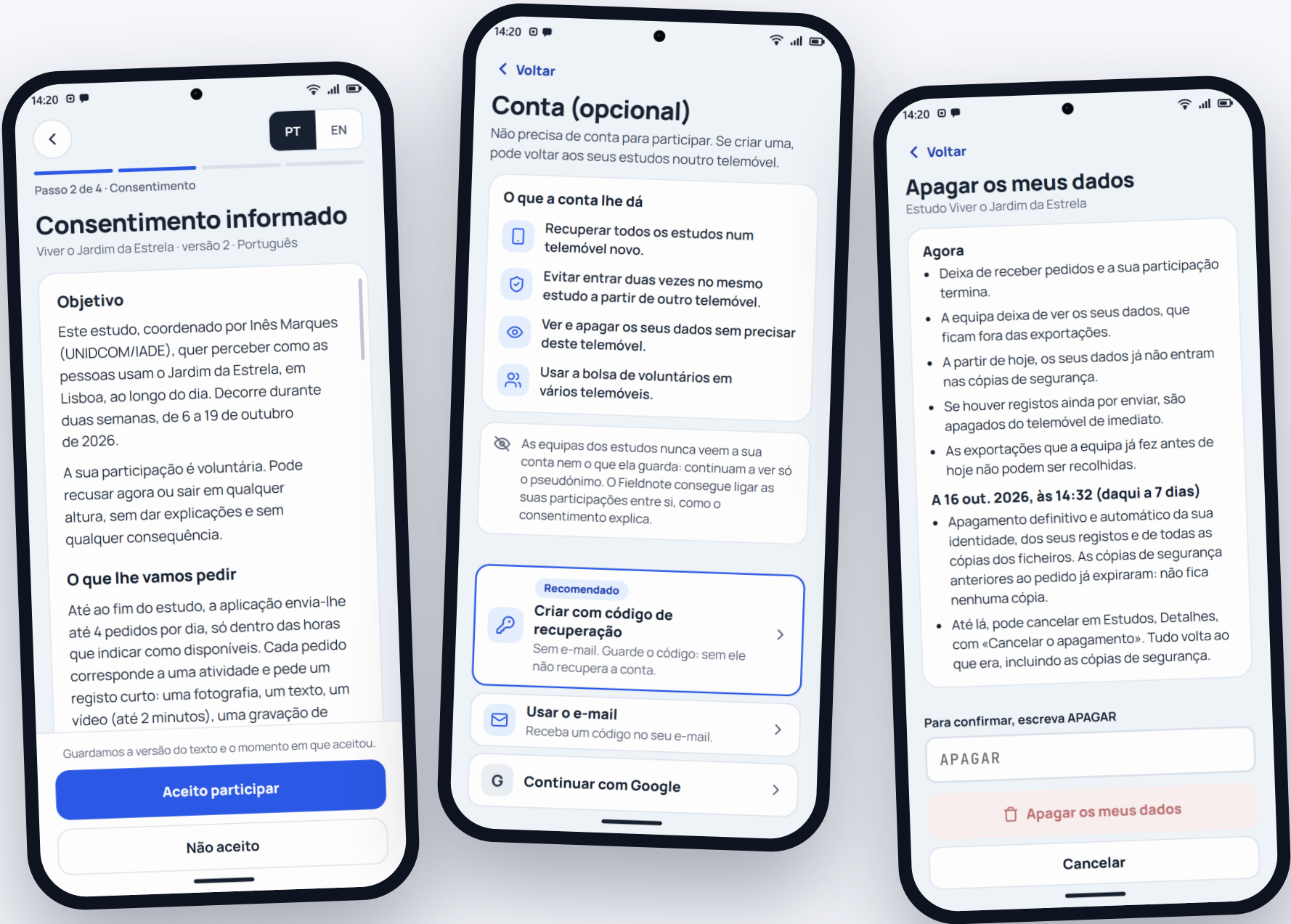
The full text is always available.
- 2

Optional account

Recovery code, email or Google, to switch phones.
- 3

Delete my data

Hidden at once, deleted on day 7.



RESEARCHER

Create the study and its activities

The study

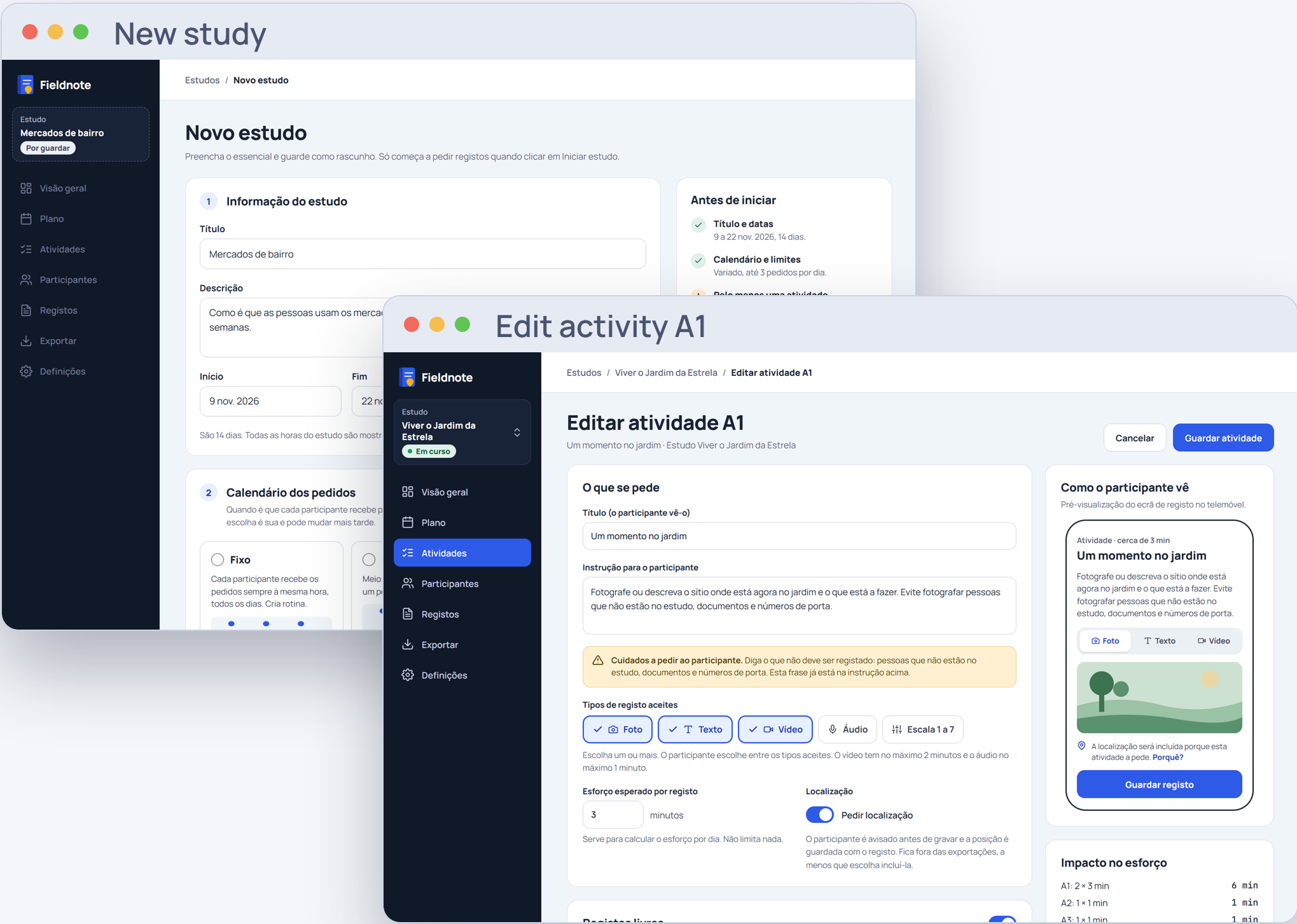
Title, dates and time zone.

The activities

What to ask for, and in which format.

The rules

When to ask, with daily limits.



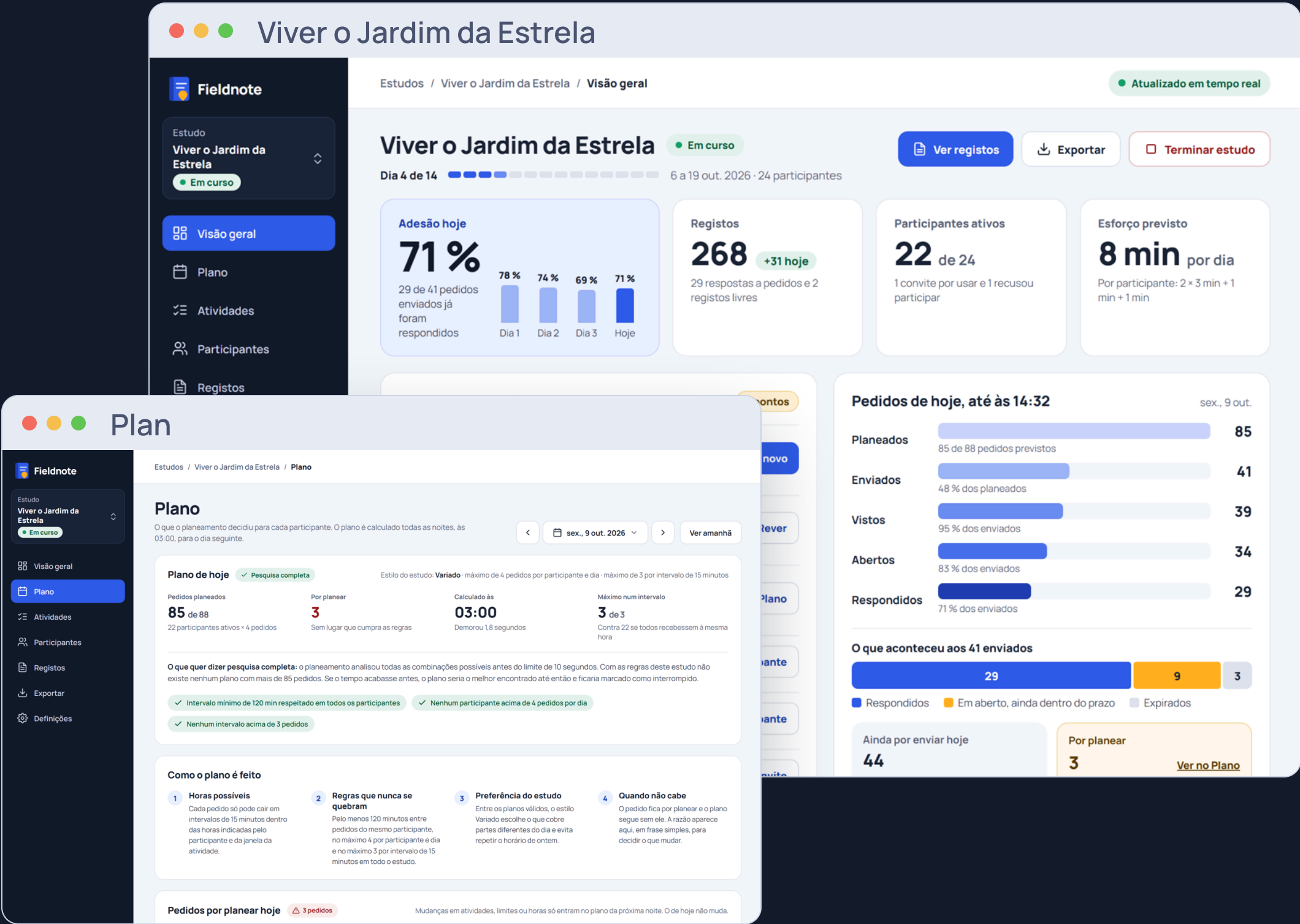
RESEARCHER

Follow the study live

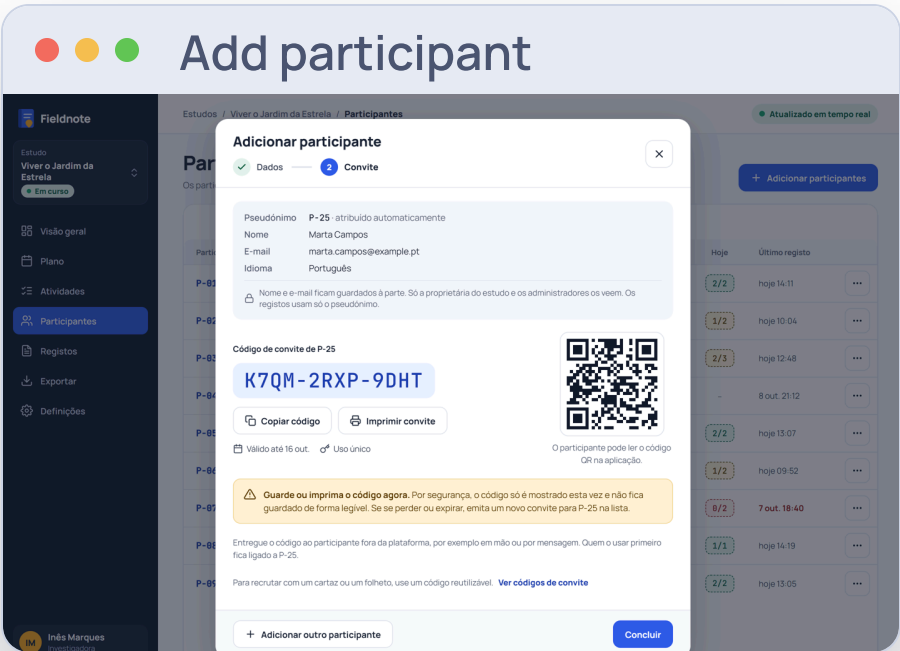
71% adherence today

268 entries received

22/24 active participants



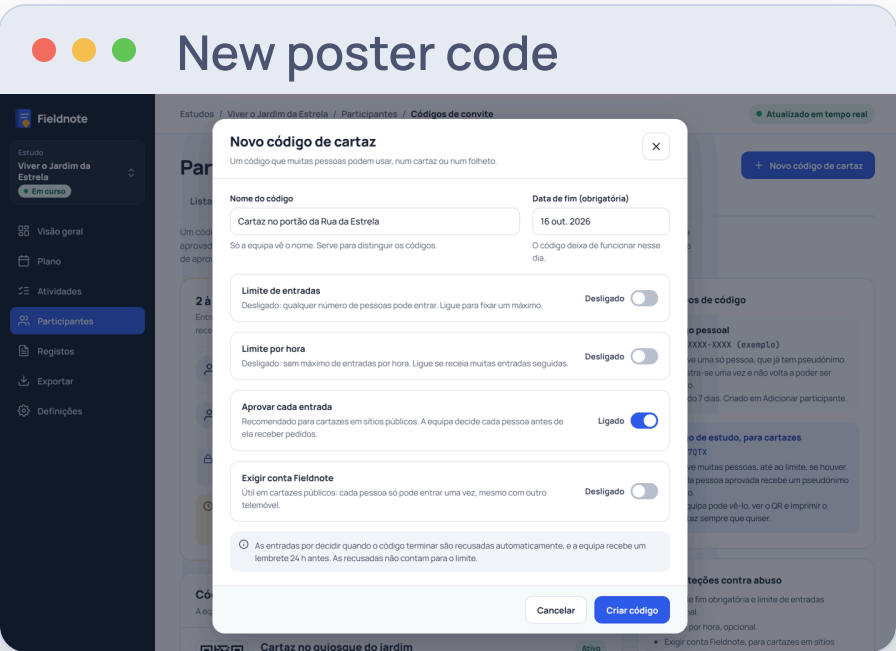
Three ways to recruit



Personal invite

A 12-character code for each person.

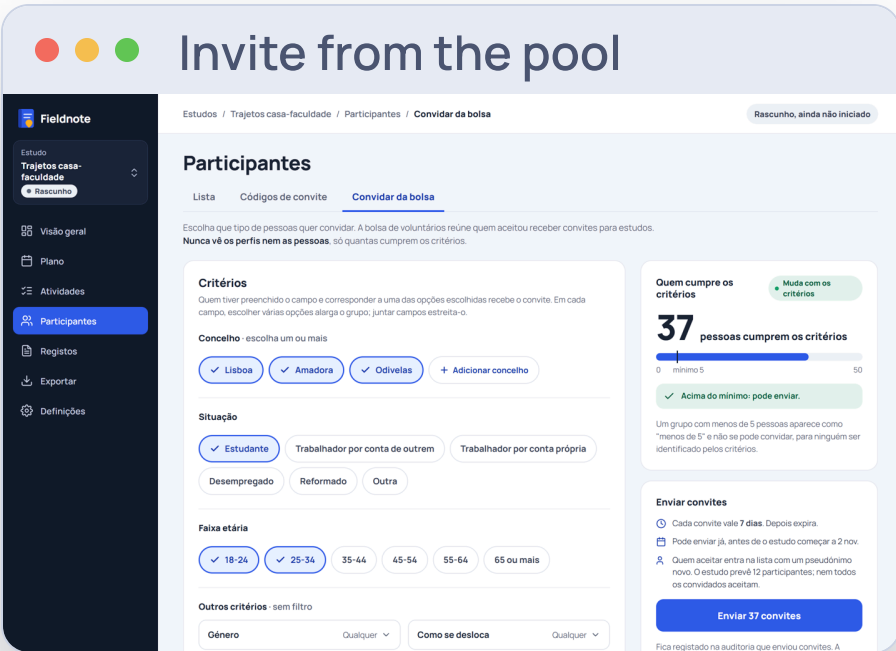
K7QM-2RXP-9DHT



Poster in the street

Reusable study code, with an entry limit and an end date.

JE4K-7QTX



Volunteer pool

The researcher only sees how many people meet the criteria.

37 people

ADMINISTRATOR

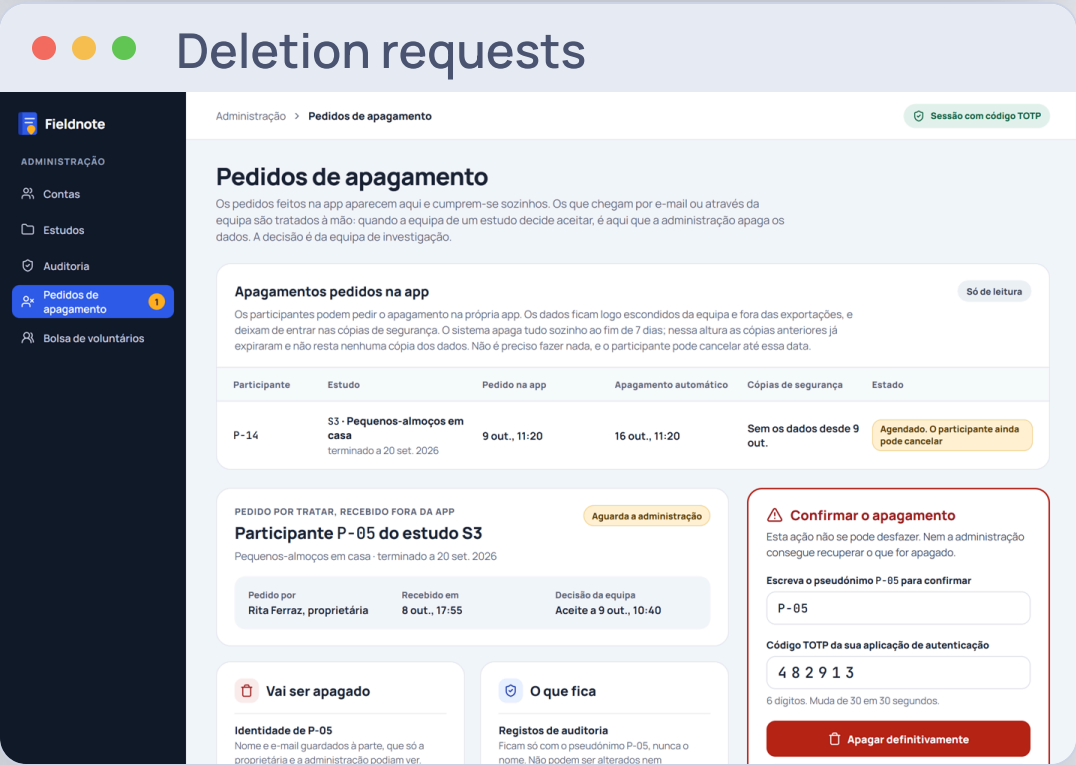
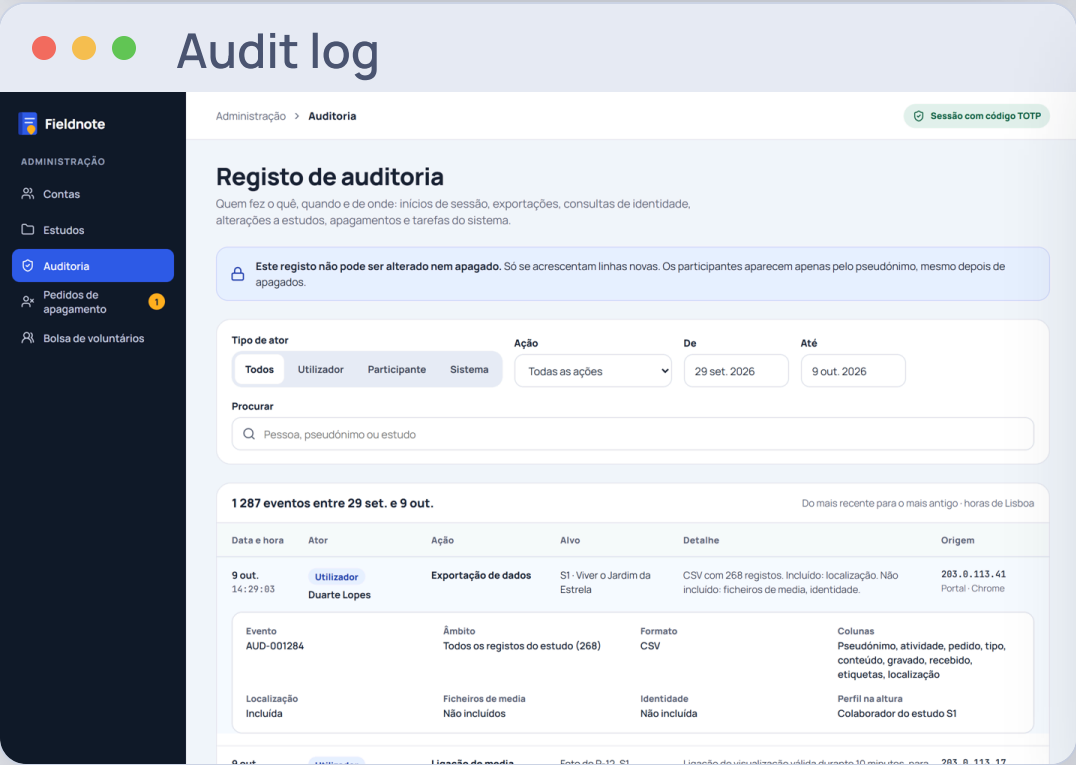
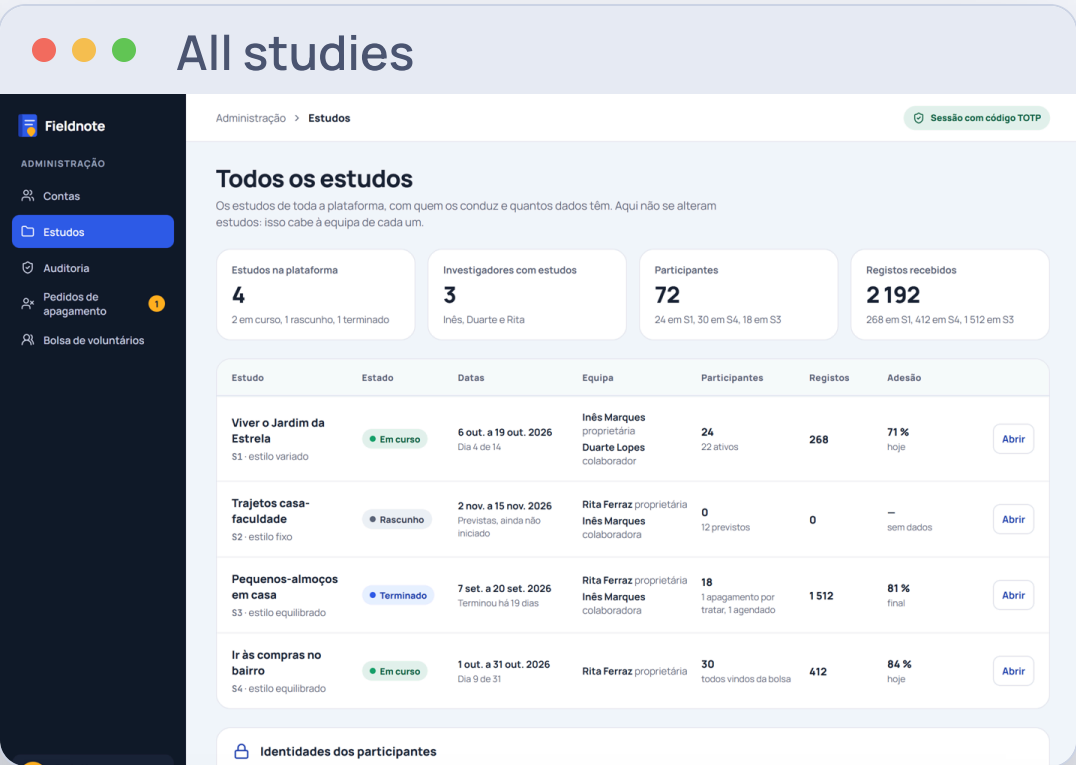
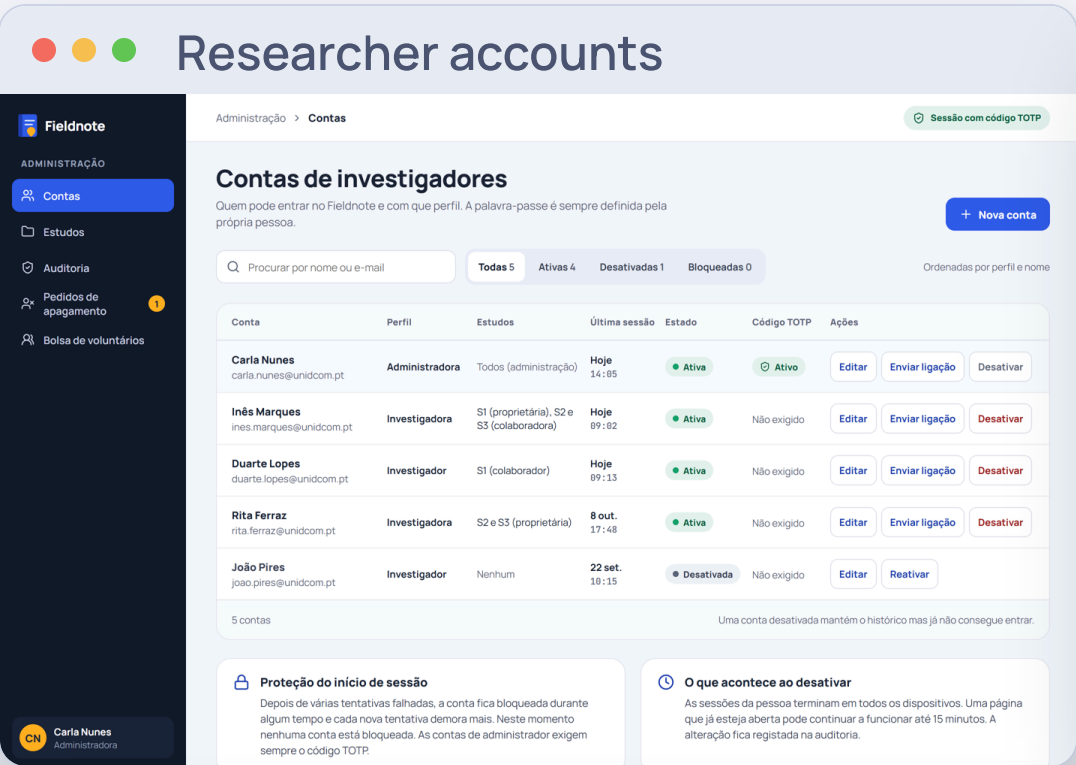
Oversee the platform

Researcher accounts, with sign-in by TOTP code

All studies on the platform

Volunteer pool

Audit log and deletion requests



ARCHITECTURE

Three nodes, one member of each cluster on each



One Docker host (Compose). Access only through the private network.

FAULT TOLERANCE

If a node goes down, the system keeps going



Node 1

Receiving entries



Node 2

Down



Node 3

Receiving entries

Keeps receiving entries if any node goes down.

A confirmed entry is never lost.

FAULTS WE WILL INJECT IN THE DEMONSTRATION

Stop containers

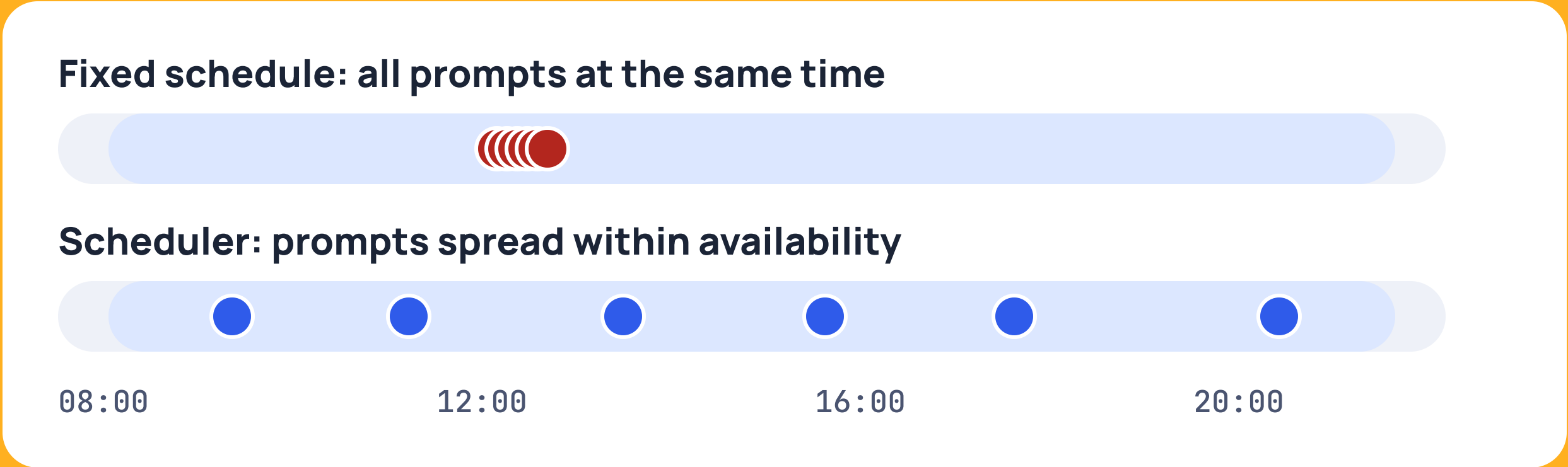
A whole node

The database primary

A storage node

Isolate a node from the network

When to ask each person for each activity



- Availability
- Activity window
- Minimum gap
- Daily limit
- Max. at once per 15 min



Approved by the AI
lecturer on 1 October
2026

SECURITY

Protect the people who take part



Authentication and roles

The administrator signs in with a TOTP code.



Protected media

Signed, short-lived links.



Audit log

Sensitive actions are logged.



Pseudonym

Participants appear as P-07.



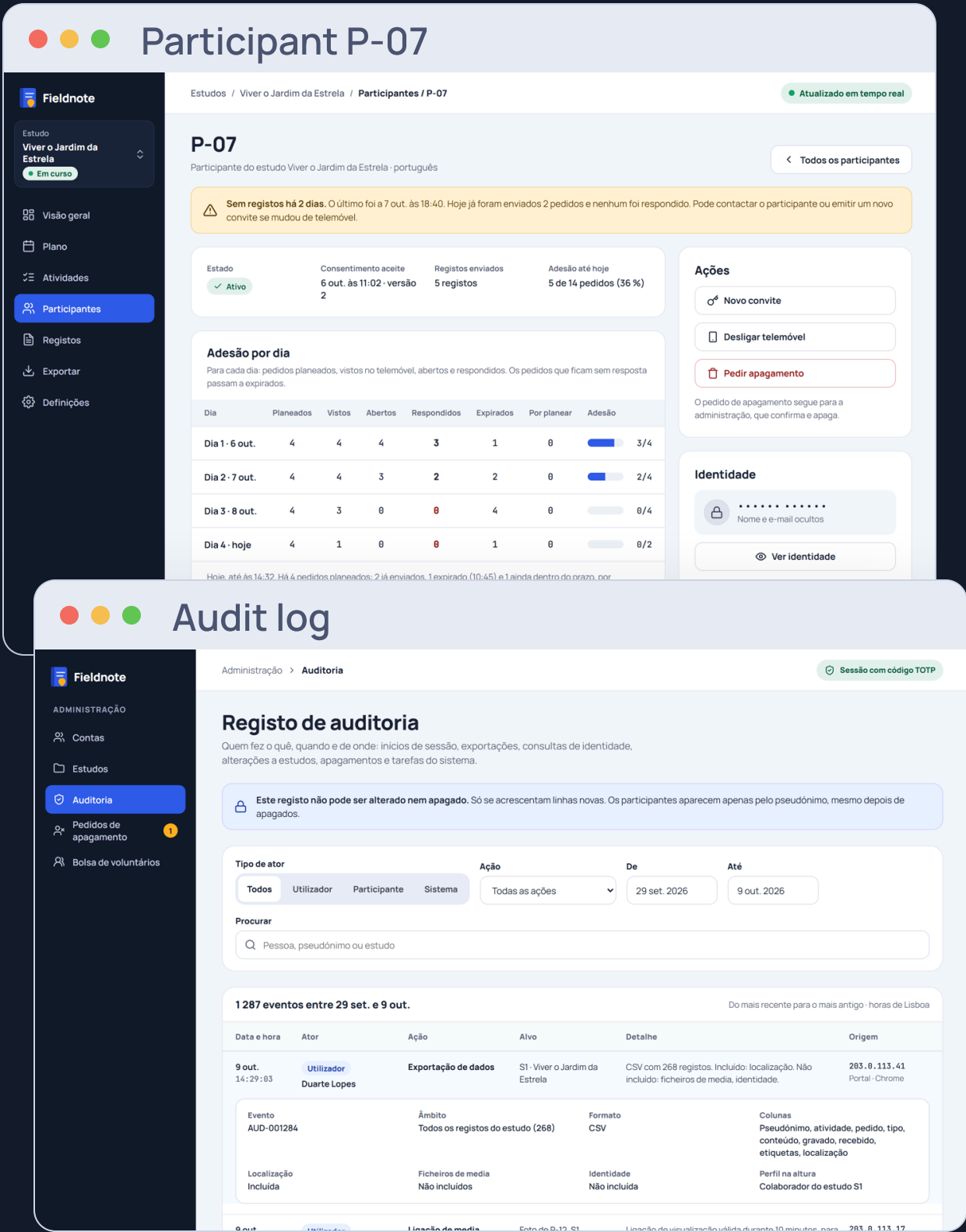
Photos without location

EXIF data is removed.



Data deletion

Fully deleted on day 7.



WHAT THE RESEARCH CHANGED

We read critically, and the design changed

- 01

Recorded time and received time

Paper diaries were filled in afterwards.
- 02

Daily limit chosen by the researcher

Two meta-analyses disagree on frequency.
- 03

Several entry formats per activity

Each person prefers one format, and recording audio in public is awkward.
- 04

Local backup notifications

Push is delivered on a best-effort basis only.

9 research files, about 21,000 words



PLAN

Four milestones to December

7 Oct

Pitch

Idea, use cases and mockups.

9 Oct

1st delivery

Design, research and fault model.

13 Nov

Prototype

Clusters running, API, app and basic portal.

18 Dec

Final delivery

Evaluated scheduler and fault demonstration.

104 tasks

GitHub Project, two-week sprints.

Approved by the lecturers

The stack and the three nodes on one host.

Division of work

Tiago: infrastructure, API, scheduler, security. César: portal, app, requirements, UML with Tiago, archive.



Questions?

github.com/Cabacadosinformaticos/fieldnote

Tiago Cabaça · César Rodrigues
IADE, Universidade Europeia · 2026-2027

